

LP_Linear_Regression

February 12, 2024

```
[7]: #Define a function that helps me estimate the house value based omn two  
    ↳parameters.  
    #1) Number of rooms  
    #2) Floor area in square meters  
    #3) A standard base price for a given area.
```

```
[8]: def estimate_house_prices(floor_area, no_of_rooms):  
    b = 10000  
    house_value = b + 16000*floor_area + 25000*no_of_rooms  
    return house_value
```

```
[11]: estimated_house_value = estimate_house_prices(9, 1)
```

```
[12]: print("Estimated House Value is: ",estimated_house_value)
```

Estimated House Value is: 179000

```
[13]: #The city authority wants a system to guide its plan  
    #approval process based on the following parameters.  
    #1) The plot area/ size of the plot  
    #2) The number of partitions/ Rooms  
    #3) The total square area in meters covered by the house from the plot  
    #4) A standard approved development fee for each zone in the city.
```

```
[15]: def city_authority_system(plot_area, no_of_rooms, sq_area):  
    b = 25000 #A standard approved development fee for each zone in the city.  
    Approved_price = b + 25000*plot_area + 1250*no_of_rooms + 3500*sq_area  
    return Approved_price
```

```
[16]: estimated_approval_costs = city_authority_system(900,6,675)  
    print("Estimated approval costs are: ",estimated_approval_costs)
```

Estimated approval costs are: 24895000

Linear Regressional Predictive problem

```
[17]: #Import necessary libraries  
    import pandas as pd  
    import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
C:\Users\ask4s\AppData\Roaming\Python\Python39\site-  
packages\pandas\core\computation\expressions.py:21: UserWarning: Pandas requires  
version '2.8.4' or newer of 'numexpr' (version '2.8.1' currently installed).
```

```
    from pandas.core.computation.check import NUMEXPR_INSTALLED
```

```
C:\Users\ask4s\AppData\Roaming\Python\Python39\site-  
packages\pandas\core\arrays\masked.py:60: UserWarning: Pandas requires version  
'1.3.6' or newer of 'bottleneck' (version '1.3.4' currently installed).
```

```
    from pandas.core import (
```

```
C:\Users\ask4s\AppData\Local\Temp\ipykernel_18476\3020193221.py:2:
```

```
DeprecationWarning:
```

```
Pyarrow will become a required dependency of pandas in the next major release of  
pandas (pandas 3.0),
```

```
(to allow more performant data types, such as the Arrow string type, and better  
interoperability with other libraries)
```

```
but was not found to be installed on your system.
```

```
If this would cause problems for you,
```

```
please provide us feedback at https://github.com/pandas-dev/pandas/issues/54466
```

```
import pandas as pd
```

```
[18]: data = pd.read_csv("D:\\MMU\\2023\\Sem 1\\NWSC\\house.csv")  
data.head()
```

```
[18]:
```

	Unnamed: 0	Area_sqm	Price_per_house
0	0	75	4.35
1	1	90	5.76
2	2	87	5.52
3	3	65	4.12
4	4	45	3.78

```
[20]: x = data.iloc[:,1]  
x
```

```
[20]:
```

0	75
1	90
2	87
3	65
4	45
5	89
6	102
7	76
8	120
9	118

```
Name: Area_sqm, dtype: int64
```

```
[22]: x = data.iloc[:,1].values.reshape(-1,1)
x
```

```
[22]: array([[ 75],
          [ 90],
          [ 87],
          [ 65],
          [ 45],
          [ 89],
          [102],
          [ 76],
          [120],
          [118]], dtype=int64)
```

```
[24]: y =data.iloc[:,2]
y
```

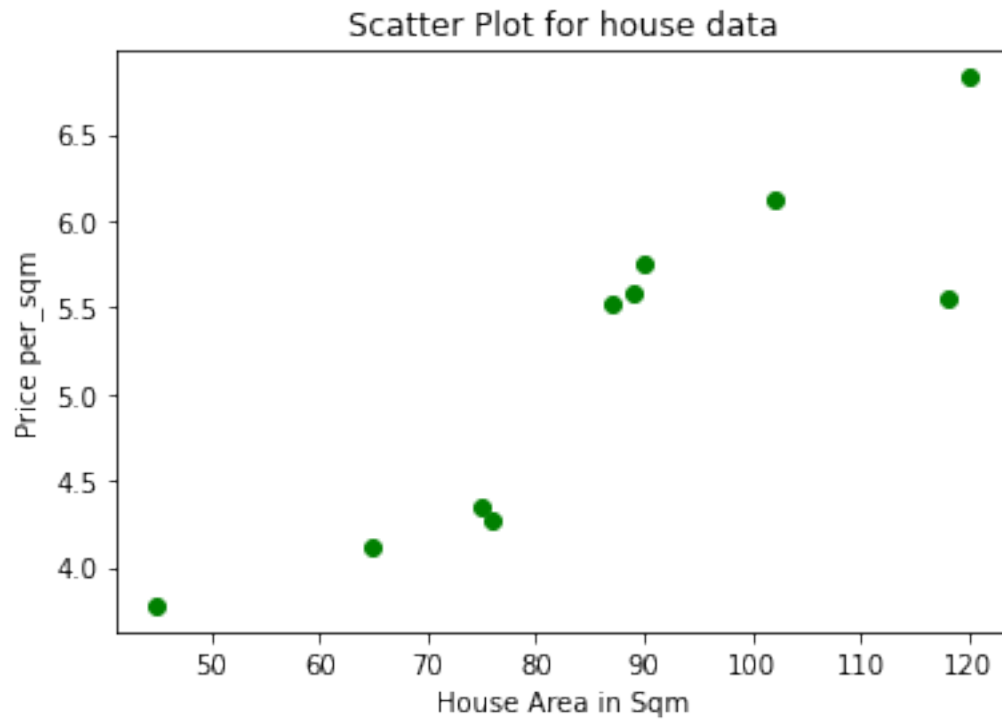
```
[24]: 0    4.35
      1    5.76
      2    5.52
      3    4.12
      4    3.78
      5    5.59
      6    6.12
      7    4.28
      8    6.83
      9    5.55
      Name: Price_per_house, dtype: float64
```

```
[25]: y =data.iloc[:,2].values
y
```

```
[25]: array([4.35, 5.76, 5.52, 4.12, 3.78, 5.59, 6.12, 4.28, 6.83, 5.55])
```

```
[28]: plt.scatter(x,y, color='Green')
      plt.xlabel("House Area in Sqm")
      plt.ylabel("Price per_sqm")
      plt.title("Scatter Plot for house data")
```

```
[28]: Text(0.5, 1.0, 'Scatter Plot for house data')
```



```
[29]: #Line of best fit is a problem handled by linear regression  
#The specific library that handles it is from sklearn
```

```
[30]: from sklearn.linear_model import LinearRegression
```

```
[31]: lr = LinearRegression()  
model = lr.fit(x,y)
```

y_Intercept

```
[32]: model.intercept_
```

```
[32]: 1.855912585154475
```

Gradient

```
[33]: model.coef_
```

```
[33]: array([0.03845545])
```

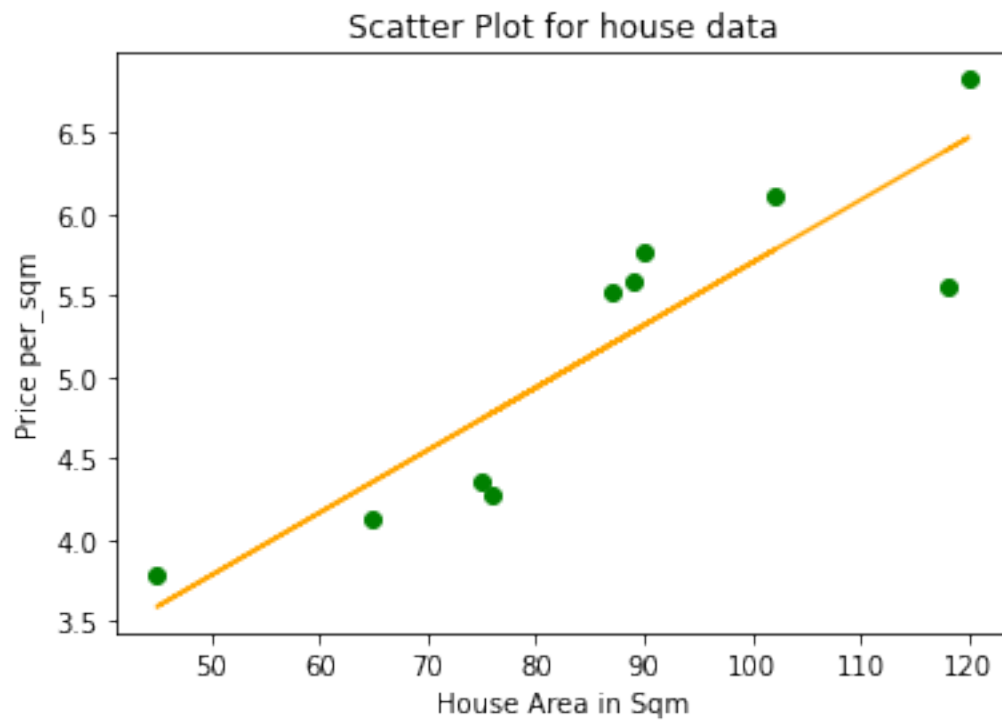
Prediction

```
[54]: y_pred = x*model.coef_ + model.intercept_  
y_pred
```

```
[54]: array([[4.74007125],  
            [5.31690298],  
            [5.20153663],  
            [4.35551676],  
            [3.58640778],  
            [5.27844753],  
            [5.77836837],  
            [4.7785267 ],  
            [6.47056645],  
            [6.39365555]])
```

```
[55]: plt.plot(x,y,'o', color='Green')  
plt.plot(x, y_pred, color = 'orange')  
plt.xlabel("House Area in Sqm")  
plt.ylabel("Price per_sqm")  
plt.title("Scatter Plot for house data")
```

```
[55]: Text(0.5, 1.0, 'Scatter Plot for house data')
```



```
[ ]:
```